Optimising NYC Taxi Operations

Course Project: EDA for NYC Yellow Taxi Data

Submitted by: Neeraj N

36M+
ANNUAL TRIPS

1.83M
SAMPLED RECORDS

263
TAXI ZONES

19
VARIABLES

This report presents an end-to-end exploratory data analysis of the NYC Yellow Taxi network for 2023, uncovering patterns in demand, pricing, and customer behaviour to guide operational strategy.

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Introduction and Problem Statement

Background, Objectives, and Analytical Approach

Introduction

New York City's yellow taxi fleet is one of the world's most recognisable urban transport systems. Despite increasing competition from ride-hailing platforms, yellow taxis continue to serve millions of passengers across the five boroughs every year. When I first looked at the scale of this dataset, 36 million trips in a single year, I knew there was a real story buried in the numbers. The 2023 trip records published by the NYC Taxi and Limousine Commission (TLC) gave us the data to tell it.

I tried to focus the analysis around three questions that actually matter if you're running or planning a taxi fleet:

- When and where is demand concentrated, and how does it vary by time of day, day of week, and season?
- How does the fare structure vary across distance tiers, vendors, hours, and passenger counts?
- What drives customer tipping behaviour, and what does it reveal about service quality perception?

> **Core Question:** What parameters (location, time of day, day of week, passenger profile) should guide a strategy to meet customer demand and optimise taxi supply across NYC?

Analytical Approach

With 36 million rows across 12 files, loading everything at once wasn't really an option on my machine. So I loaded each month separately, grouped trips by date and pickup hour, and sampled 5% from each group. Stratifying by hour was important so that quiet slots like 3 AM in January weren't just dropped from the sample entirely. All numbers in this report are based on that cleaned sample, and where I've given full-year estimates I've scaled them back up using the 5% fraction.

02

Data Preparation and Cleaning

Sampling Strategy, Column Fixes, Missing Values and Outlier Removal

Sampling Strategy

Loading all 12 files together wasn't feasible, so each month was loaded and sampled separately. Trips were grouped by date and pickup hour, and 5% were taken from each group. The reason for stratifying by hour rather than just random sampling was to make sure off-peak slots like 3 AM in January were still represented and not just lost in the noise.

~36M	5%	1,896,400	1,830,747
FULL DATASET ROWS	SAMPLING FRACTION	ROWS AFTER SAMPLING	ROWS AFTER CLEANING

Column Fixes

The raw dataset contained two airport fee columns due to a schema inconsistency across monthly files. These were merged using combine_first() to retain non-null values. The store_and_fwd_flag column was dropped as it carried no analytical value. Three derived columns were added: pickup_hour, pickup_month, and pickup_day_of_week.

Missing Value Treatment

Column	Missing Count	Pct of Data	Treatment Applied	Reasoning
passenger_count	64,874	3.42%	Mode imputation (1.0)	Solo travel is most common
RatecodeID	64,874	3.42%	Mode imputation (1.0)	Standard rate code is the norm
congestion_surcharge	64,874	3.42%	Filled with 0	Missing implies no surcharge applied
airport_fee	64,874	3.42%	Filled with 0	Missing implies non-airport trip

Additionally, 29,681 rows with passenger_count equal to zero were replaced with the mode value of 1, as a trip with zero passengers is operationally invalid.

Outlier Removal

Negative values in monetary columns were converted to absolute values, treating them as data entry errors. The following outlier filters were then applied sequentially:

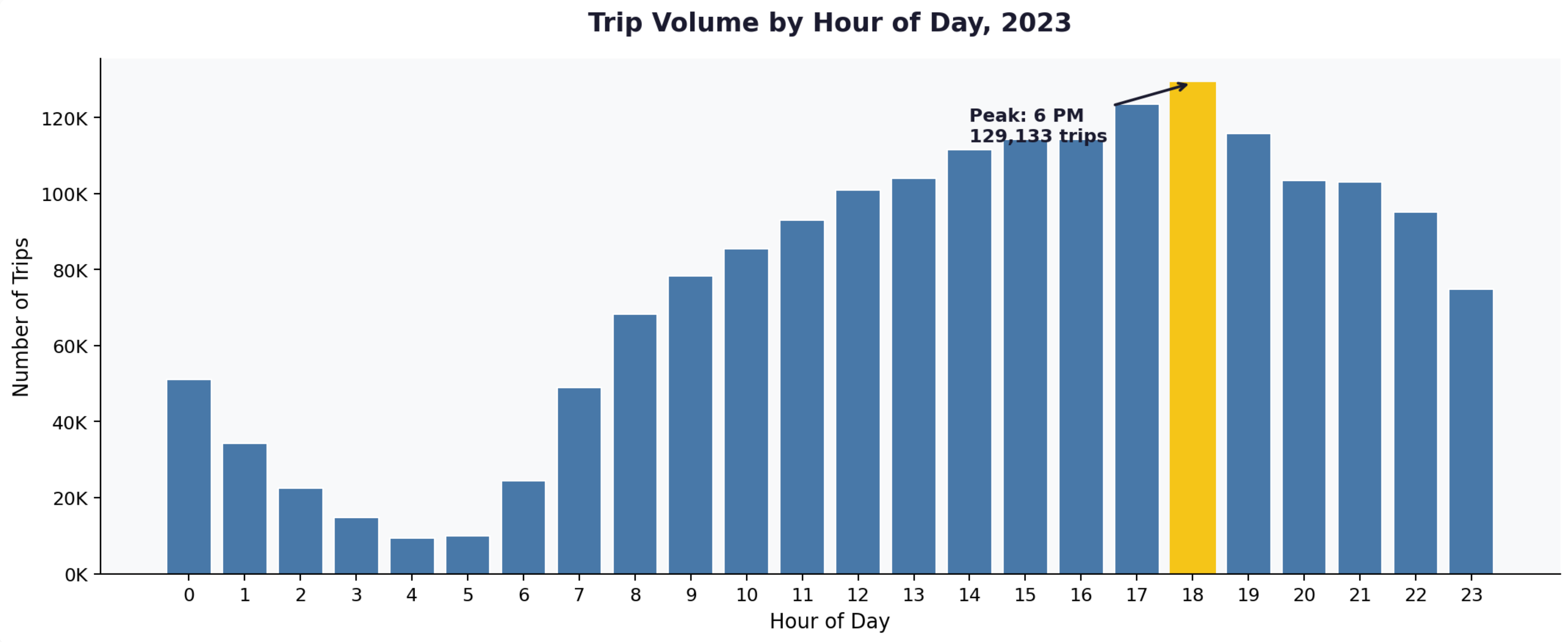
Outlier Type	Filter Condition	Rows Removed	Justification
Excess passengers	passenger_count greater than 6	21	NYC legal maximum capacity
Extreme distance	trip_distance greater than 250 miles	46	No NYC taxi trip exceeds 250 miles
Impossible fare	distance under 0.1 mi AND fare over $100	682	Data entry error pattern
Ghost trips	distance = 0, fare = 0, different zones	63	Trip recorded but no movement
Invalid payment	payment_type equals 0	64,874	Not a recognised payment code
Extreme fare	fare_amount greater than $1,000	2	No valid NYC taxi fare at this level

03

General EDA

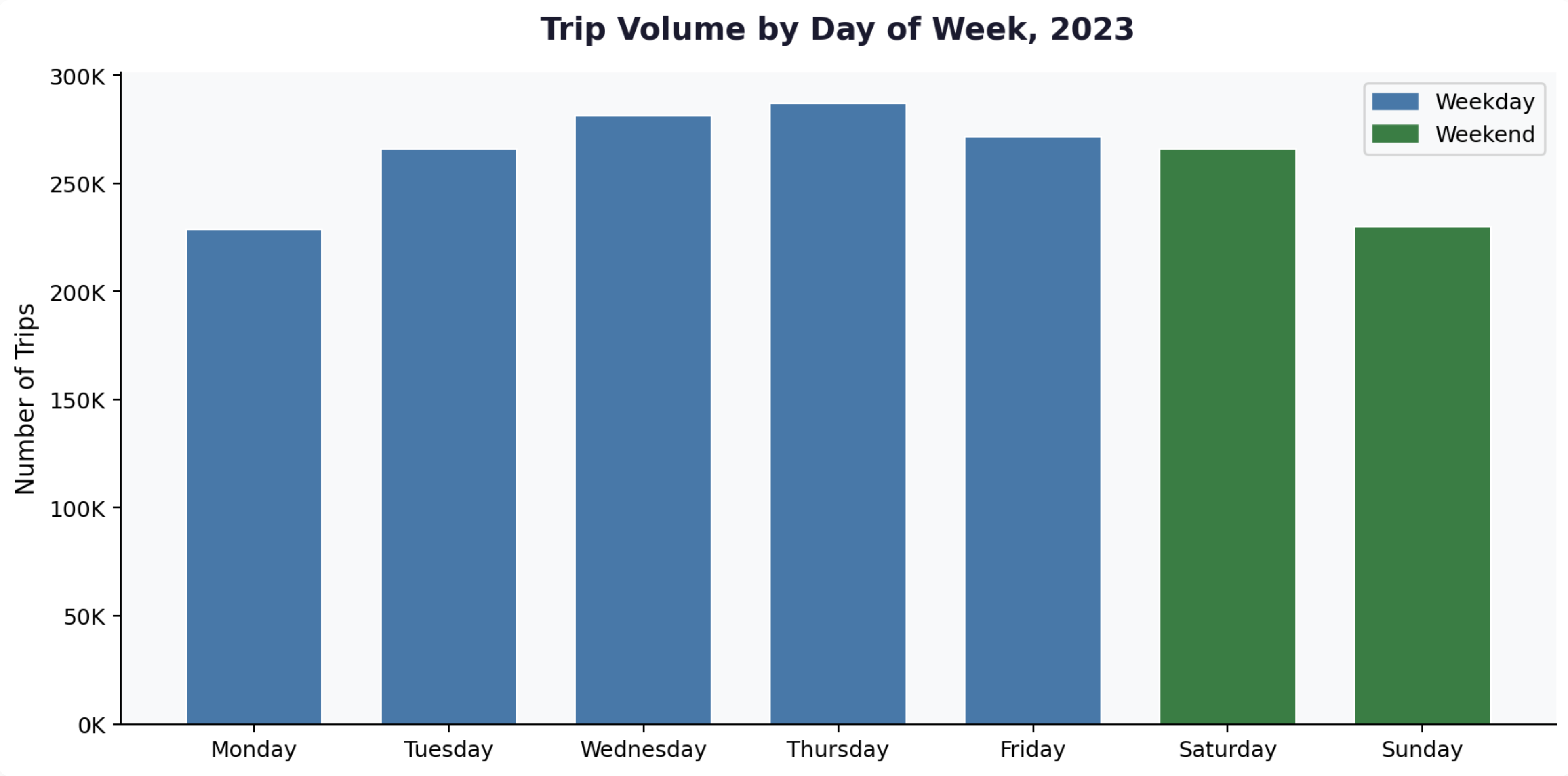
Temporal Patterns, Financial Analysis, Payment Behaviour and Geography

Temporal Patterns



Trip volume by hour of day. The yellow bar marks the peak at 6 PM with 129,133 sampled trips, equivalent to approximately 2.56 million annual trips in that single hour window.

> Demand bottoms out between 4 and 5 AM, then climbs steadily through the morning. The sharp spike at 6 PM is the evening commuter rush in action. What surprised me was how consistant this shape is across all 12 months, whether it's January or July, the peak barely shifts.



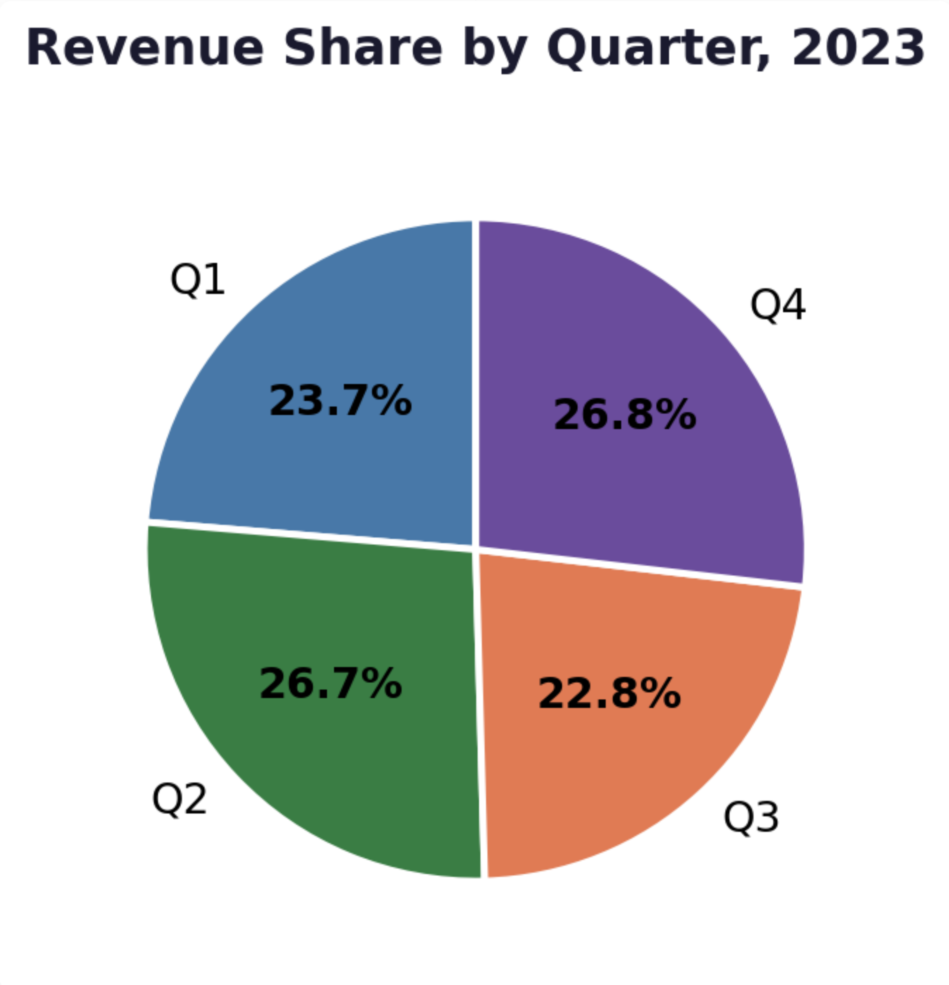
Trip volume by day of week. Thursday is the busiest weekday. Weekend demand is notably lower in total volume but follows a distinct pattern.

Thursday being the busiest day was a bit unexpected. But more interesting is how weekends don't just have fewer trips, the whole shape of demand changes. Instead of that sharp evening spike you get a flatter curve that runs from midday all the way through to late evening.

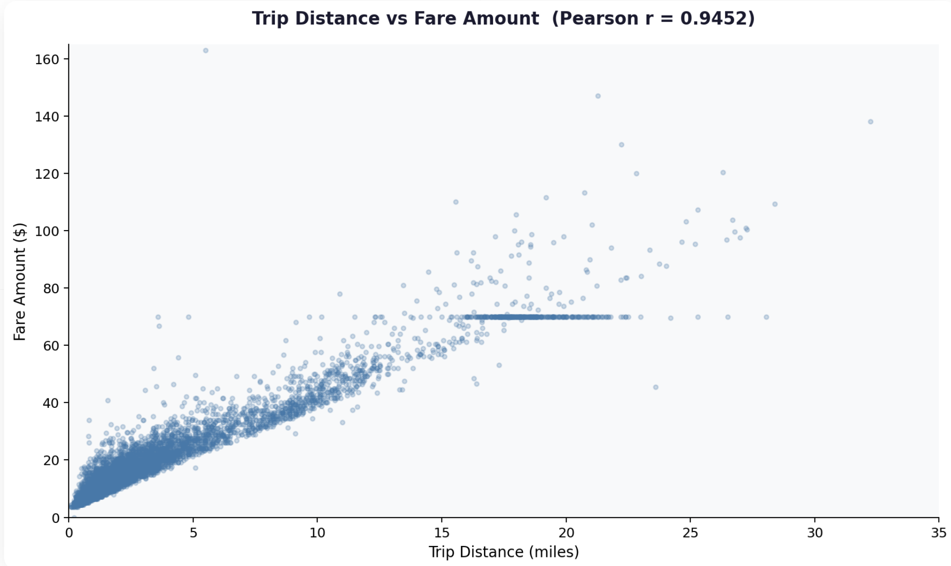
Monthly trip volume across 2023. May and October are the two peak months. The summer dip (July to September) is consistent with reduced business travel and tourist season patterns.

Demand wasn't flat across the year. May and October stand out as the clear peaks — a mix of spring and autumn tourism plus business travel. The summer dip from July through September was actually one of the more unexpected findings. That's when tourist demand should be high, but it doesn't fully compensate for the drop in regular commuter traffic. December recovering strongly suggests end-of-year activity plays a role too.

Financial Analysis

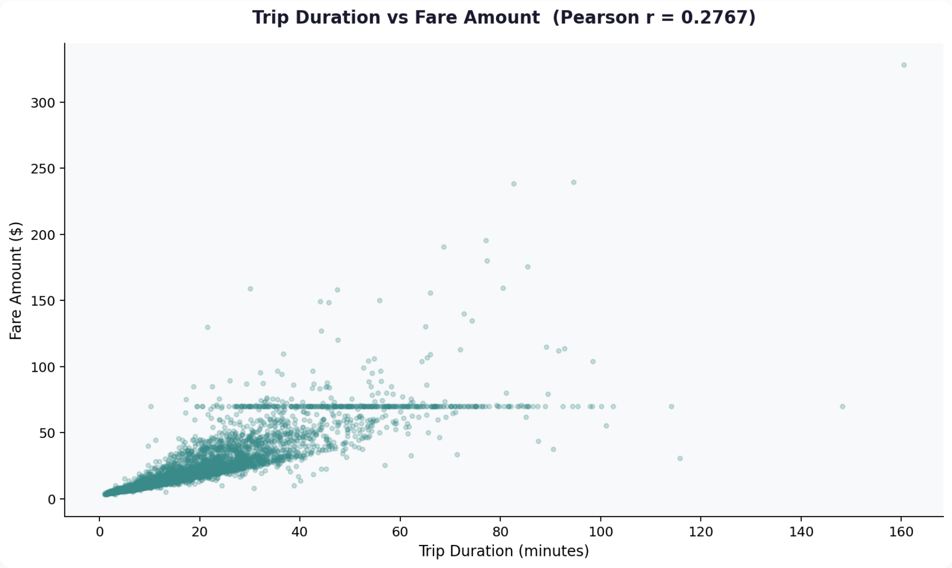


Q4 and Q2 each generate approximately 26.8% of annual revenue, while Q3 is the weakest quarter at 22.7%.

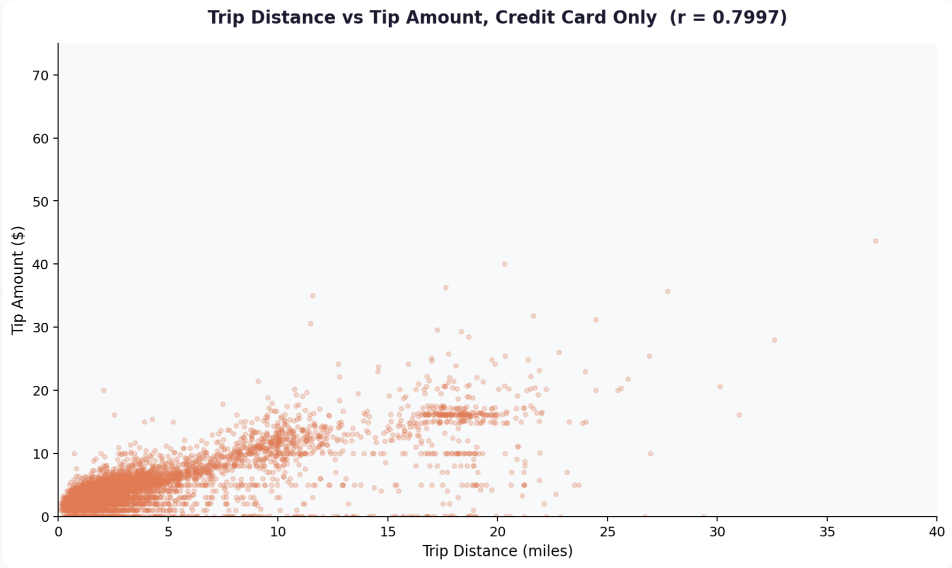


Strong linear relationship between trip distance and fare amount (r = 0.9452). The horizontal band at $70 represents JFK flat-rate trips.

> An r of 0.9452 between distance and fare is about as clean as real-world data gets. If you look at the scatter plot there's also a flat horizontal band of dots sitting at $70 for longer trips. That's the JFK flat rate showing up, which is a nice visual confirmation that the data is real and not just noise.

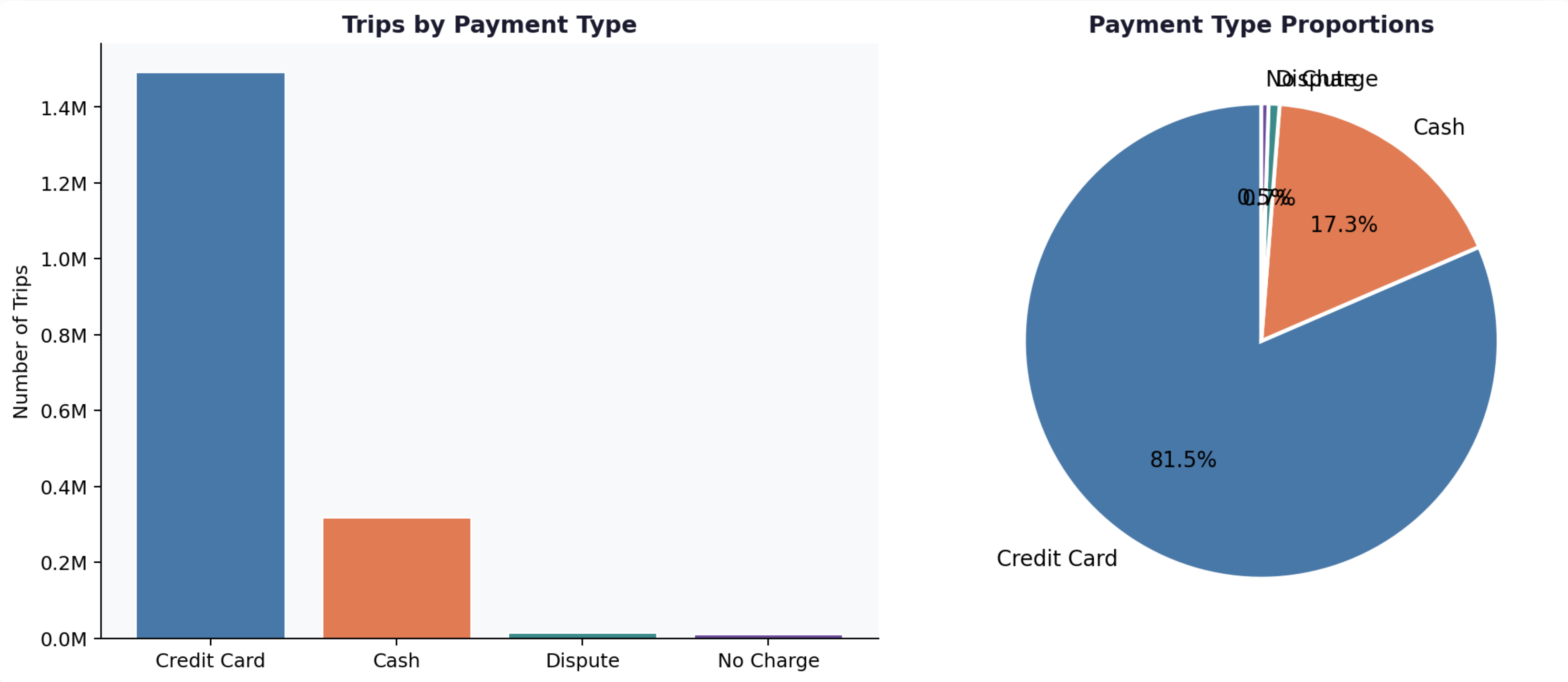


Trip duration vs fare amount (r = 0.2767). Duration has a much weaker relationship with fare than distance, meaning traffic congestion does not directly increase fare revenue.



Trip distance vs tip amount for credit card payments only (r = 0.7997). Cash tip data is systematically absent from the dataset.

Payment Behaviour



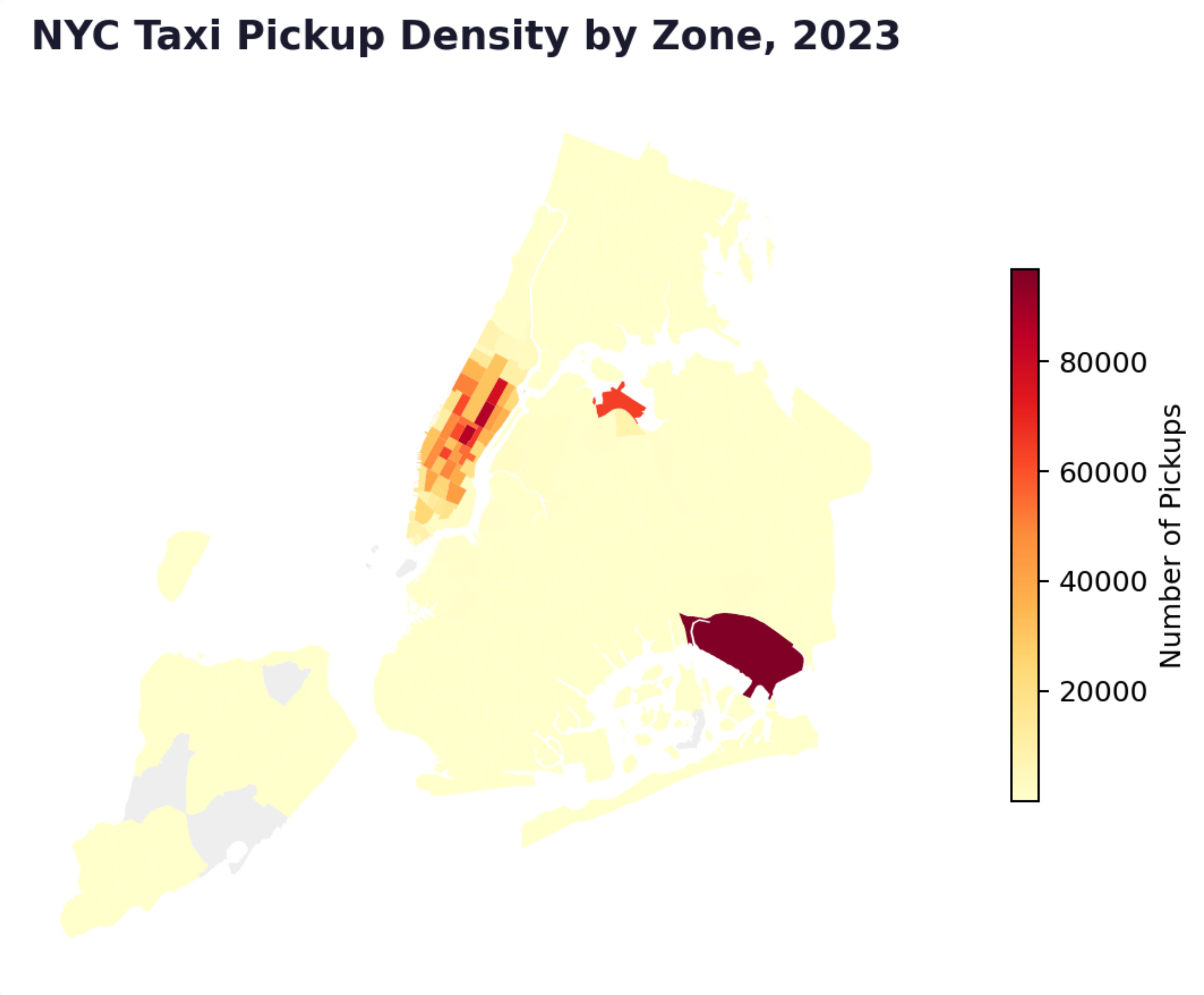
Payment type distribution. Credit card dominates at 81.5% of all trips. Cash accounts for 17.3%. Dispute and No Charge categories are negligible.

Over 81% of trips go through card which is actually really useful for the analysis. The catch is that cash tips aren't captured in the dataset at all, so everything in the tip section is based only on card transactions. It's worth keeping that in mind when reading those numbers.

The full breakdown by payment method confirms how card-dominant the NYC yellow taxi system is. Cash still accounts for roughly 1 in 6 trips, but everything else, disputes and no-charge trips, makes up less than 1.5% combined.

Payment Type	Trip Count	Share of Total
Credit Card	1,491,760	81.5%
Cash	316,336	17.3%
Dispute	13,651	0.75%
No Charge	9,000	0.49%

Geographic Analysis



NYC taxi pickup density by zone. The two dark hotspots are JFK Airport (Zone 132, bottom right) and the Midtown/Upper East Side Manhattan corridor (centre strip).

> Manhattan is basically the entire map here. Upper East Side and Midtown form this dense band in the middle and almost everything else is light. JFK is the one exception, sitting all the way out in Queens with the highest raw pickup count of any single zone. And the outer boroughs barely register at all, which felt like a big gap once I saw it visually.

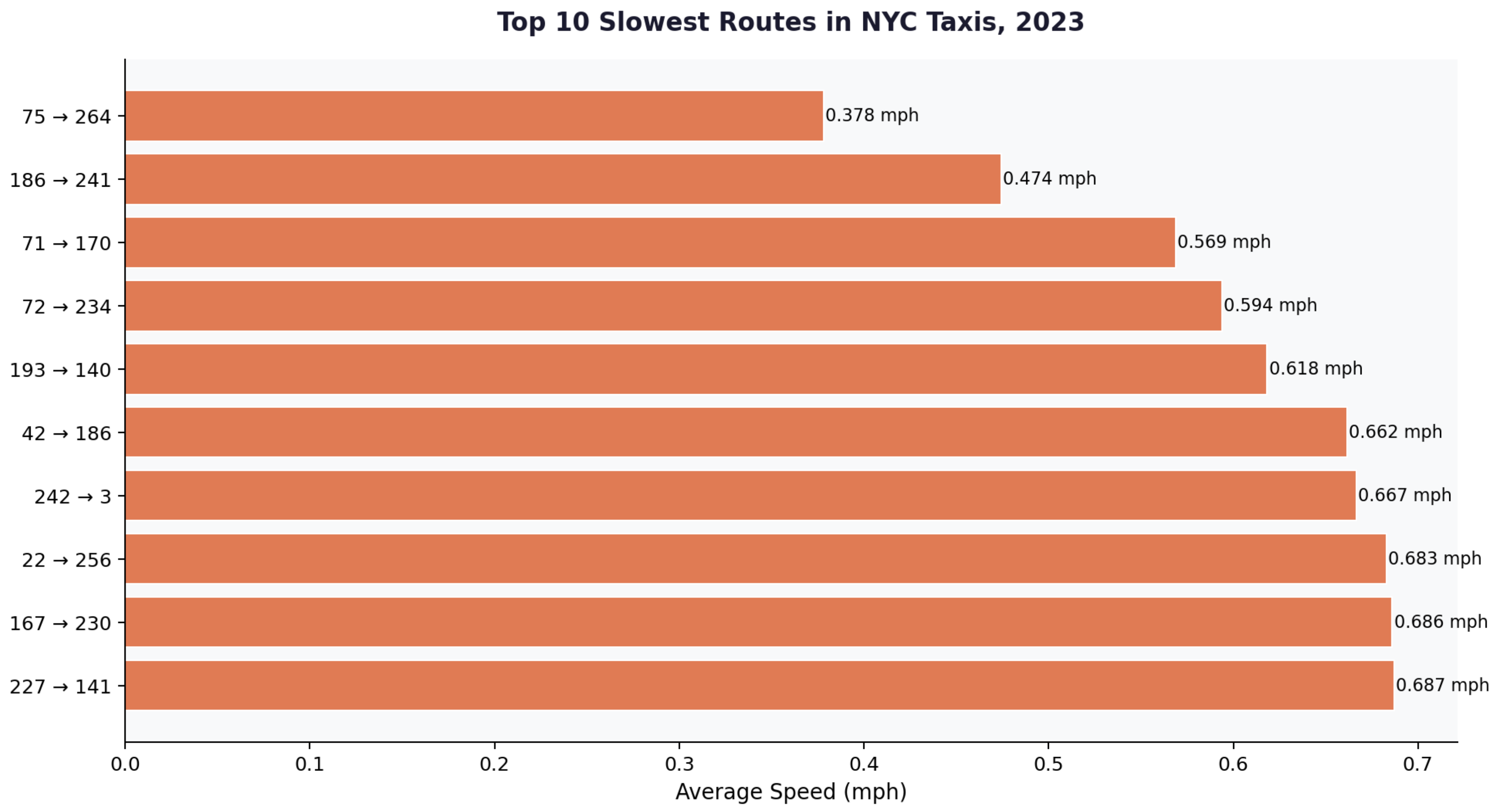
Rank	Zone ID	Area	Sampled Pickups
1	132	JFK Airport	94,940
2	237	Upper East Side South	86,473
3	161	Midtown Center	85,400
4	236	Upper East Side North	77,120
5	162	Midtown East	65,210

04

Detailed EDA

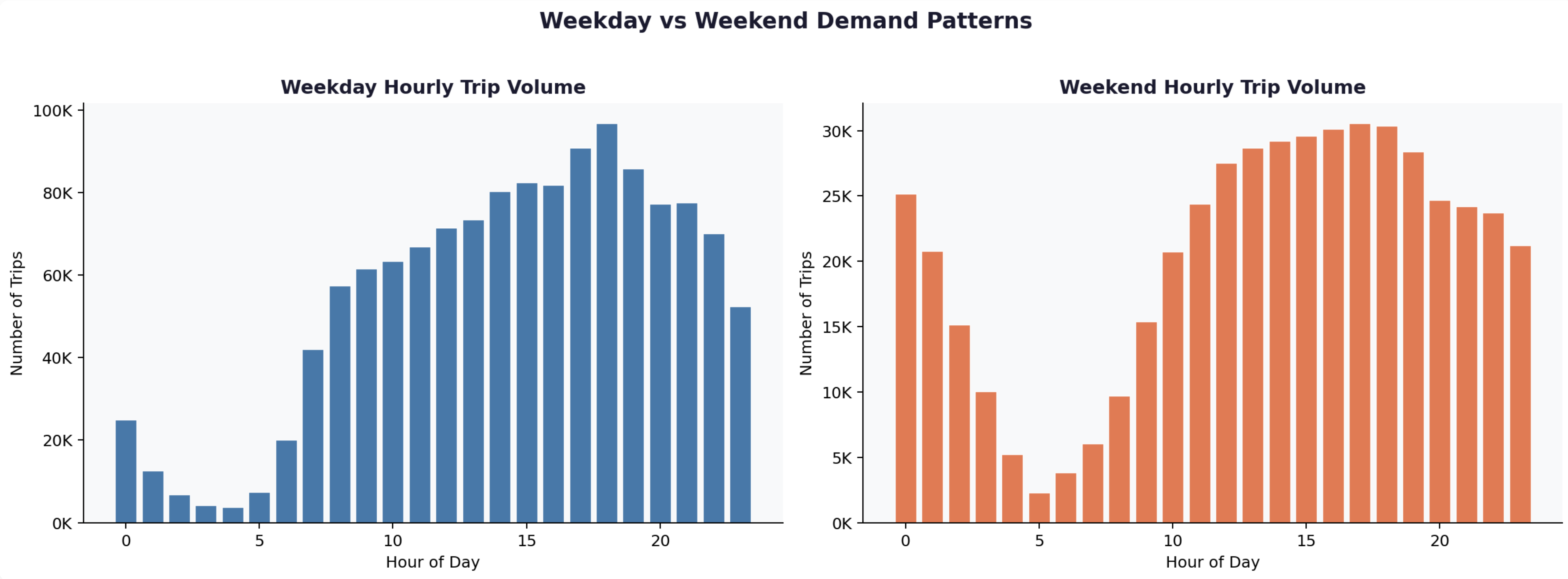
Operational Efficiency, Pricing Strategy and Customer Experience

Operational Efficiency



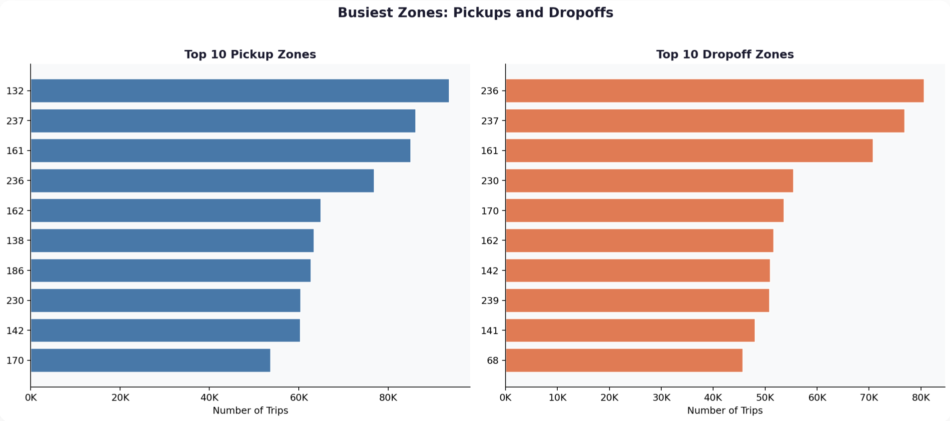
The 10 slowest routes in 2023, measured by average speed in mph. All involve very short distances (0.1 miles) with very long durations, concentrated during midday hours.

I expected the slowest routes to be long cross-city trips stuck in rush hour traffic but they're actually tiny short hops of about 0.1 miles that somehow take over an hour. And they're mostly happening between noon and 3 PM, not at 6 PM. Midday congestion in some of these zones is genuinely brutal in a way the hourly chart doesn't fully capture.

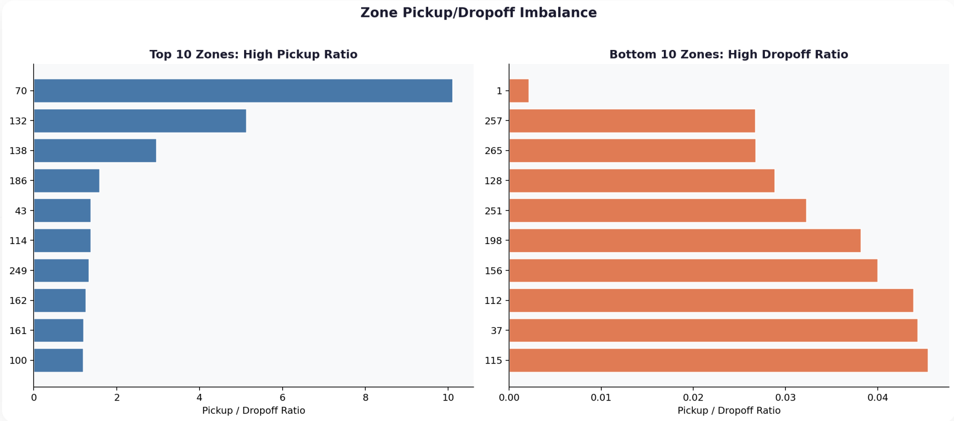


Weekday hourly demand (blue) shows a sharp single peak at 6 PM driven by commuters. Weekend demand (orange) is lower in volume but flatter and broader, peaking between noon and 8 PM.

These two charts look nothing like each other. Weekdays are all about that 6 PM spike from commuters. Weekends build slowly through the afternoon, stay elevated through the evening, and don't have a single sharp peak. It's basically two different operating modes and probably needs two different approaches to where you put your cabs.

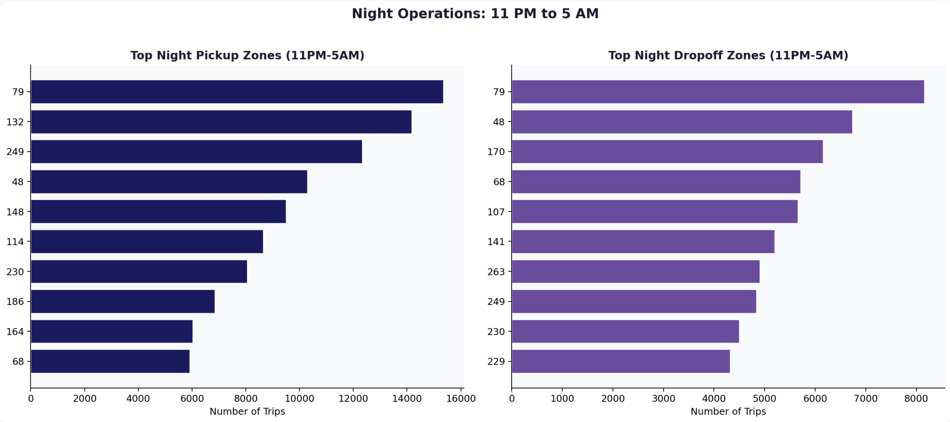


Top 10 pickup zones (blue) versus top 10 dropoff zones (orange). JFK Airport tops pickups but does not appear in the top dropoff list.

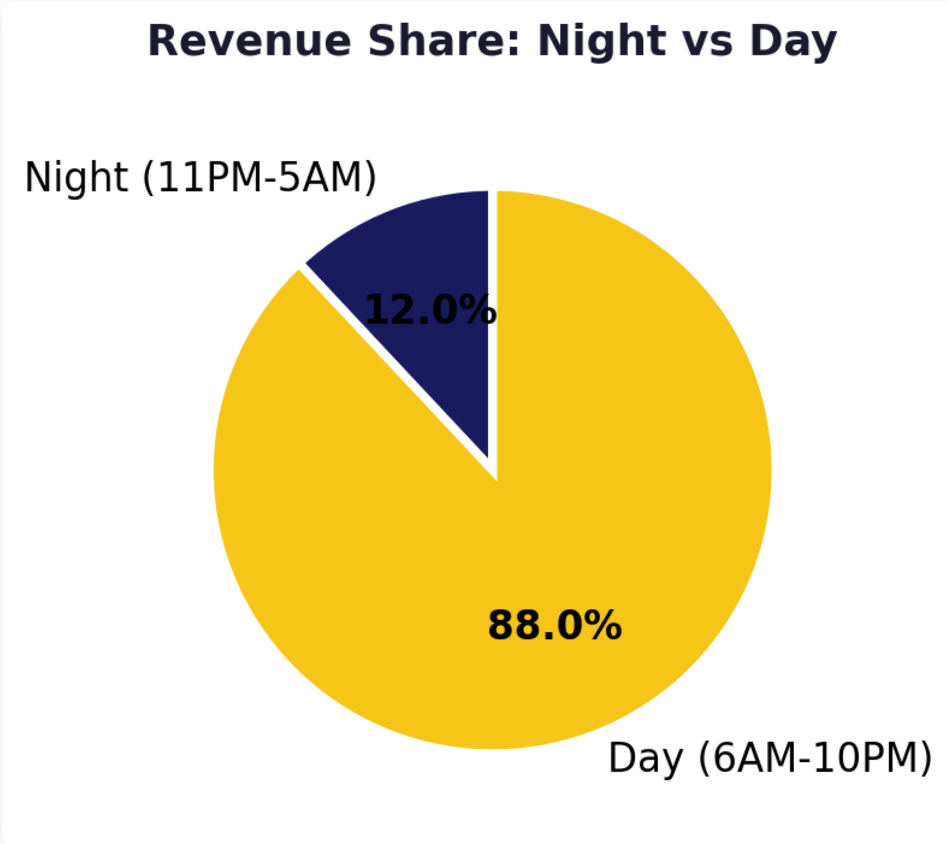


Pickup to dropoff ratio by zone. Zone 70 has a ratio of 9.8 and JFK (Zone 132) has 5.06, meaning far more cabs arrive for pickups than for dropoffs. Zone 1 (Newark Airport) is the reverse at 0.003.

> This one genuinely surprised me when the numbers came out. JFK is a 5:1 pickup to dropoff ratio, so cabs flood in for pickups and then drive back to the city empty. Newark is the complete opposite and barely gets any pickups at all. Nobody seems to have figured out a good way to balance this and it's clearly costing drivers on every airport run.



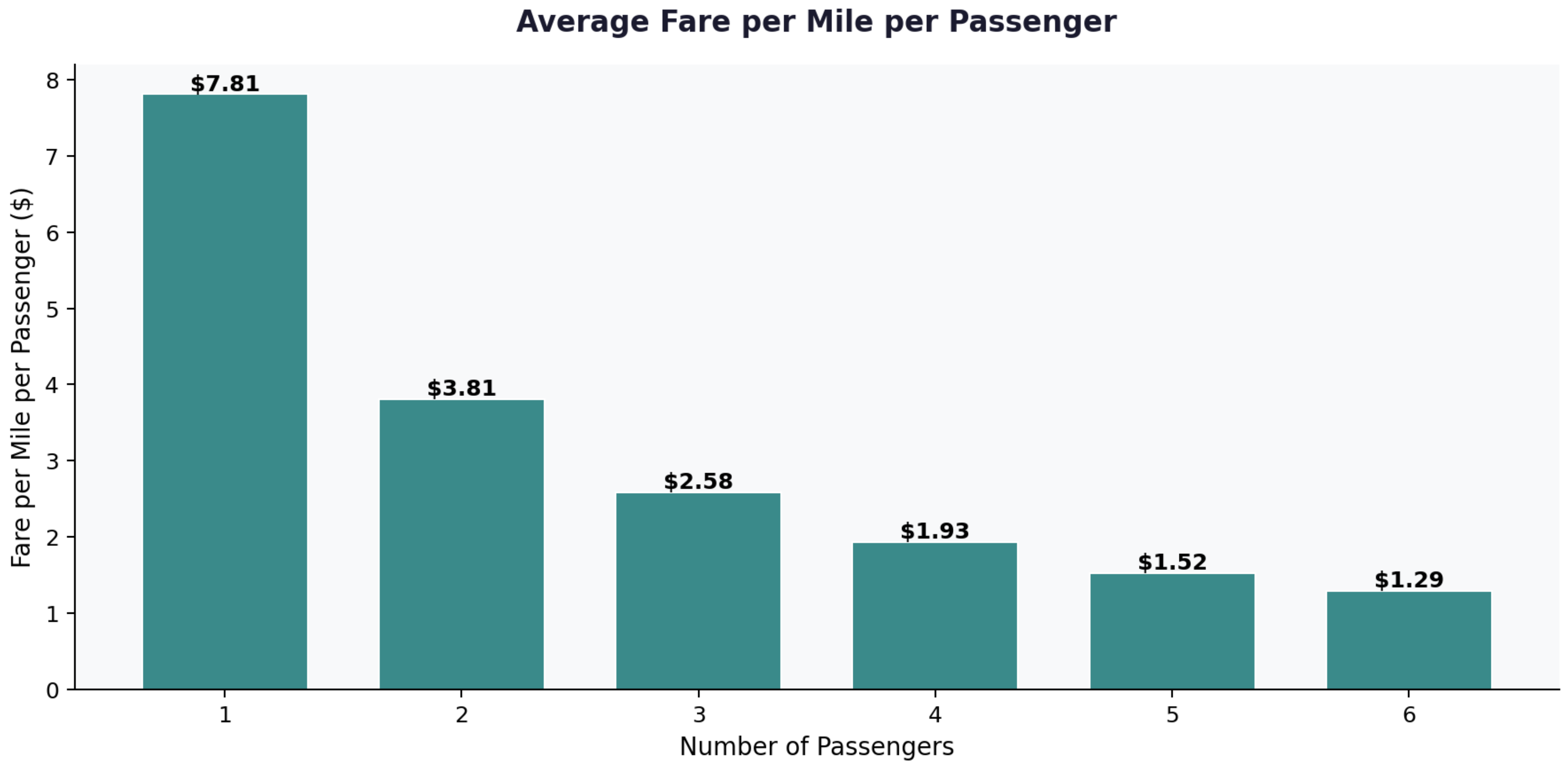
Top night pickup and dropoff zones between 11 PM and 5 AM. Zone 79 (East Village) leads both charts, followed by Zone 132 (JFK Airport) for pickups.



Night hours (11 PM to 5 AM) generate just 12% of total revenue despite being a distinct demand segment driven by nightlife and late-arriving flights.

East Village topping both night pickup and dropoff charts makes complete sense if you know the neighbourhood. The more interesting number is the 12% revenue share for all night hours. For a segment that's clearly active, 12% feels low, which suggests there probably isn't enough supply out there to match the demand after midnight.

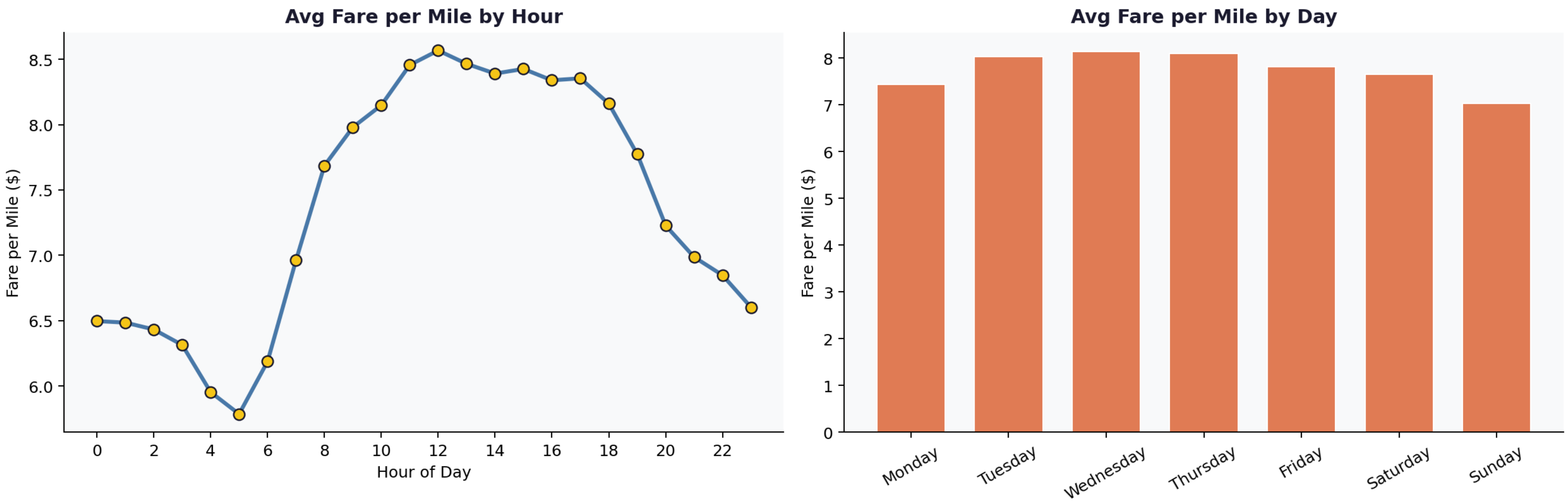
Pricing Analysis



Average fare per mile per passenger. Solo riders pay $7.81 per mile individually. Six-passenger groups pay only $1.29 per mile per person.

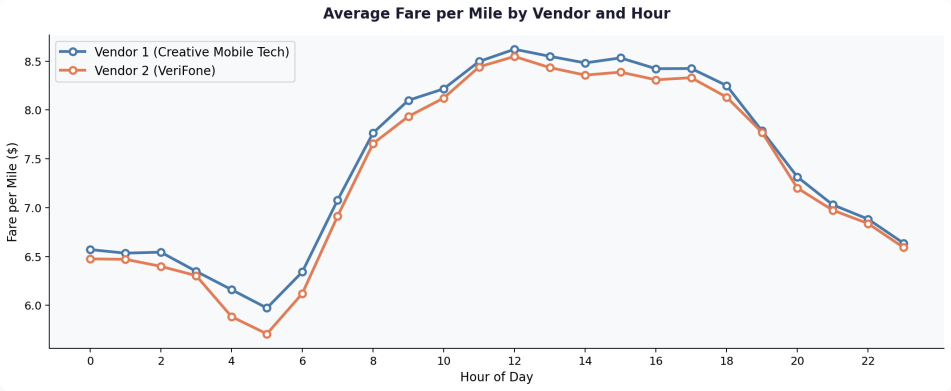
The drop in per-person cost for groups is bigger than I expected. A solo rider pays about $7.81 per mile while a group of six is splitting $1.29 per mile each. That's over 80% cheaper per person. Groups are getting a really good deal and probably don't realise it, which is an opportunity if you wanted to market to them.

Fare per Mile: Temporal Patterns

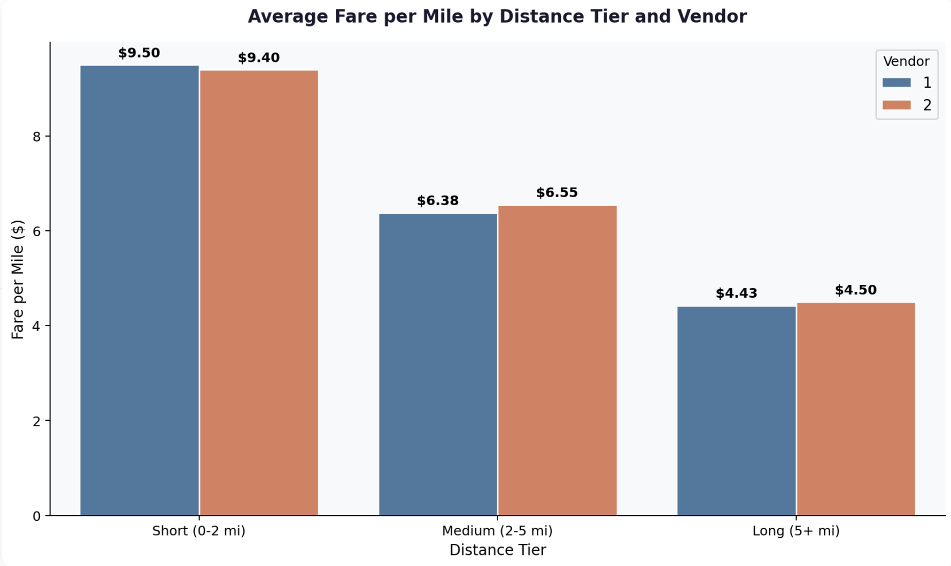


Fare per mile peaks at noon ($8.56/mi) and is lowest at 5 AM ($5.80/mi). Weekdays charge more per mile than weekends, with Sunday being the cheapest day overall.

Midday having the highest per-mile rate actually makes sense once you think about it. Lunch trips are short, and short trips all get the same base fare regardless of distance, so the rate per mile ends up higher. Sunday is the cheapest day per mile because people are taking longer rides that spread that base fare over more distance.

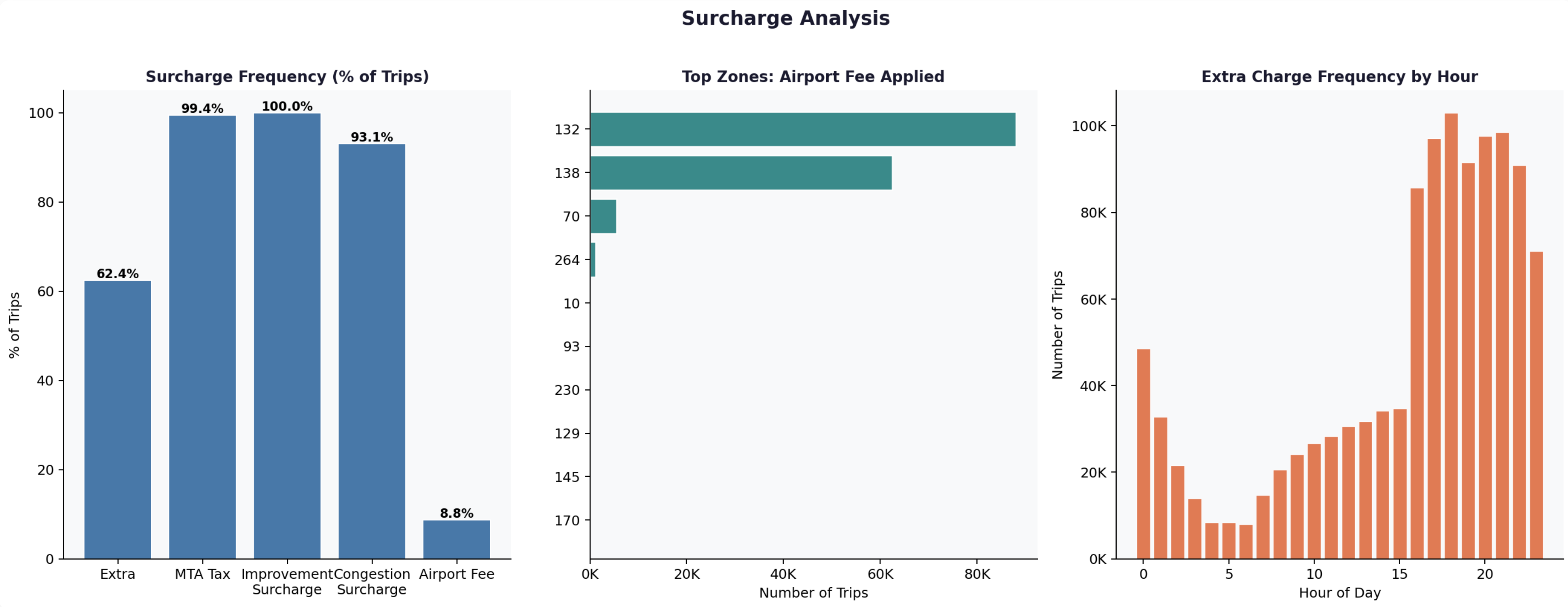


Vendor 1 (Creative Mobile Technologies) charges marginally more per mile than Vendor 2 (VeriFone) across most hours, but the gap is negligible throughout the day.



Fare per mile by distance tier and vendor. Both vendors show nearly identical pricing: short trips ~$9.50/mi, medium ~$6.50/mi, long ~$4.45/mi.

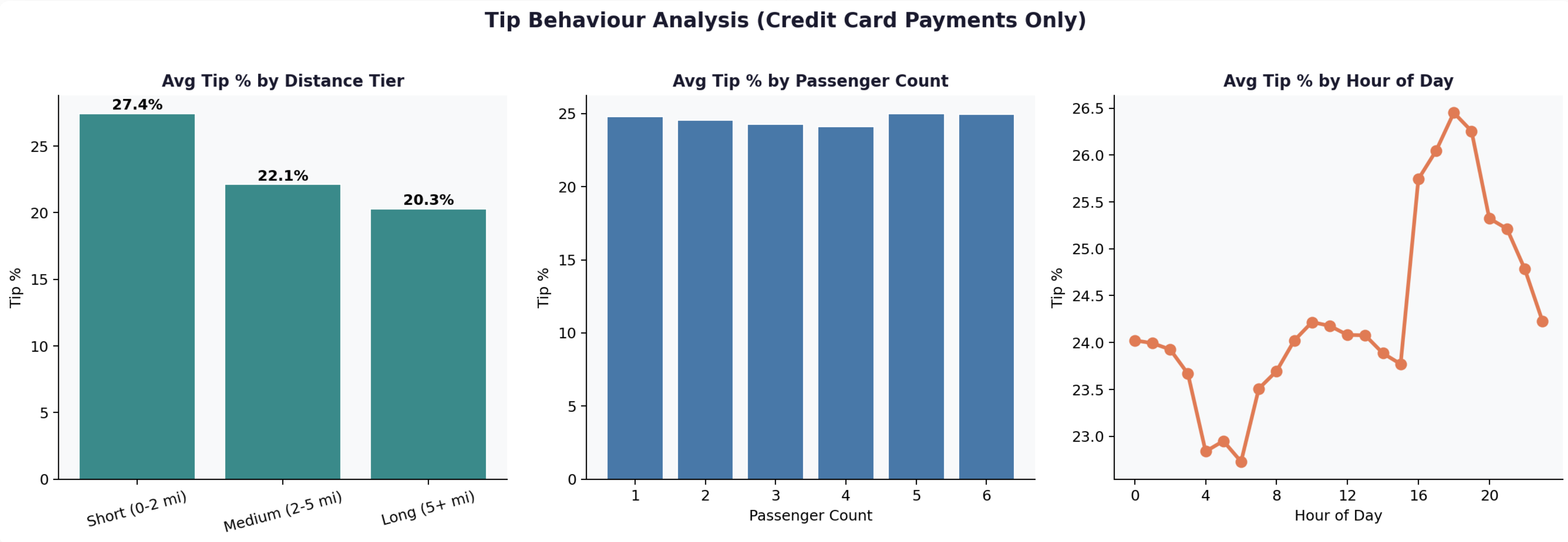
The two vendors are charging essentially the same rates at every hour and for every trip type. There's no real pricing competition happening here. The difference between short and long trip rates isn't either vendor doing anything strategic, it's just the fixed base fare spreading over more miles on longer trips. Both vendors have basically the same math.



Surcharge frequency analysis. MTA Tax and Improvement Surcharge apply to nearly every trip. Airport Fee is rare at 8.77%, concentrated almost entirely at JFK and LaGuardia.

Surcharge	Frequency	Interpretation
Improvement Surcharge	99.96%	Universal, applied on virtually every trip
MTA Tax	99.44%	Universal, applied on virtually every trip
Congestion Surcharge	93.07%	Near-universal for Manhattan trips
Extra (rush/overnight)	62.41%	Applied during peak and overnight windows
Airport Fee	8.77%	Concentrated at JFK and LaGuardia only

Customer Experience

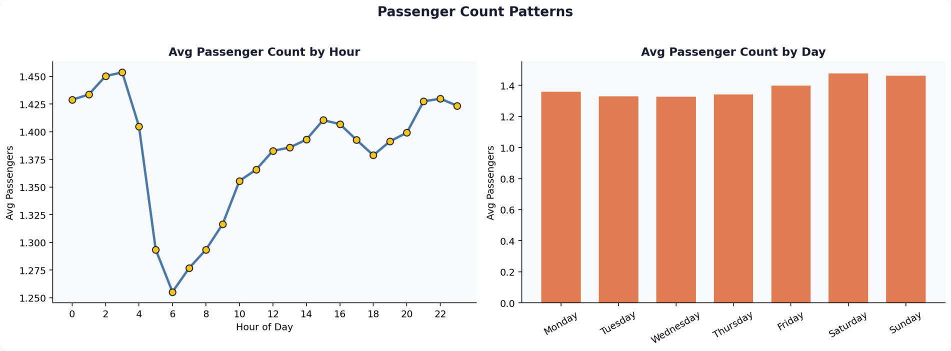


Tip percentage analysis across three dimensions. Short trips receive the highest tips (27.4%). Passenger count has minimal effect. Tips peak between 6 and 7 PM.

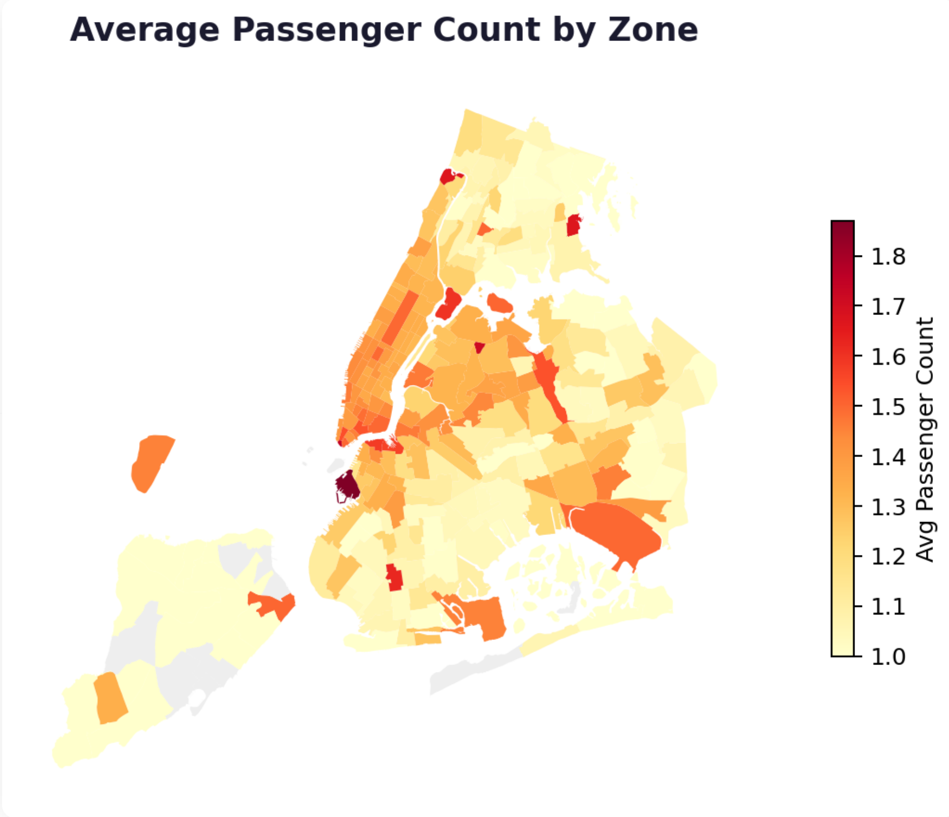
> This was probably the most interesting finding in the whole analysis. Short trips get tipped the most, percentage-wise. On a $14 fare, 25% is just $3.50. On a $25 fare it starts to feel steep. People tip by feel, not by formula, and that shows up clearly in the data.

Comparing the profile of low and high tippers shows the pattern clearly. High tippers take shorter, cheaper trips; low tippers take longer, more expensive ones. Passenger count barely differs between the two groups, which rules out group dynamics as an explanation.

Metric	Low Tippers (under 10%)	High Tippers (over 25%)
Average Trip Distance	4.88 miles	2.30 miles
Average Fare	$25.56	$14.41
Average Passenger Count	1.38	1.36
Average Duration	21.4 minutes	12.3 minutes



Average passenger count by hour (left) and by day (right). Late night hours (2 to 4 AM) carry the most passengers per trip. Weekends average slightly more passengers than weekdays.



Average passenger count by zone. Darker zones in upper Manhattan and outer areas carry more passengers per trip. Central Manhattan zones are lighter, reflecting predominantly solo commuter trips.

The late-night passenger spike at 2 to 4 AM makes sense when you think about who's travelling then. People aren't commuting alone at that hour, they're heading home with friends. The outer zones also tend to carry more passengers per trip overall which fits with those being more leisure and social trips compared to the solo commuter pattern in central Manhattan.

05

Conclusions and Recommendations

Routing Efficiency, Zone Positioning and Pricing Strategy

4.1.1 Routing and Dispatching Optimisation

Demand peaks at 6 PM on weekdays and Thursday is consistently the busiest day. The congestion problem is actually worst not at 6 PM but during the midday window from noon to 3 PM, where some routes are clocking average speeds below 0.1 mph. These aren't long cross-city trips in traffic, they're tiny short hops that just aren't going anywhere.

RECOMMENDATIONS

- Pre-position cabs in Midtown and Upper East Side zones by 4:30 PM to capture the evening rush before demand reaches its peak at 6 PM.
- De-prioritise short inner-city dispatches during 12 PM to 3 PM when congestion brings average speeds to near-zero on several routes. Redirect drivers to faster outer-borough or highway-adjacent routes.
- At night (11 PM to 5 AM), dispatch should favour Zone 79 (East Village) and Zone 132 (JFK Airport), which top both the night pickup and dropoff charts across the full year.
- Early morning hours (2 AM to 4 AM) carry more passengers per trip than any other window. Group ride options during this period could improve revenue per trip without adding fleet pressure.

4.1.2 Strategic Zone Positioning

Manhattan handles the bulk of pickups and dropoffs, but the airport zones are where the big inefficiency is. JFK has a 5:1 pickup to dropoff ratio, so drivers queue up for pickups and then drive back to the city without a passenger. Newark is the exact opposite and barely gets any pickups. Neither situation is working well for driver earnings.

RECOMMENDATIONS

- Deploy dedicated airport queues at JFK and LaGuardia during morning and evening flight peaks to reduce idle waiting time and improve driver utilisation per shift.
- Incentivise drivers dropping off at low-pickup zones (Newark, Zone 265) to reposition quickly to high-demand Manhattan zones rather than waiting locally for a pickup that may not come.
- On weekends, shift supply southward toward entertainment districts in Lower Manhattan and Midtown from noon onward, as leisure demand builds through the afternoon and peaks later than weekday patterns.
- Outer boroughs, particularly the Bronx and Staten Island, are consistantly underserved. A targeted zone coverage strategy could open a new demand segment without cannibalising existing Manhattan volume.

4.1.3 Pricing Strategy and Revenue Optimisation

Both vendors are charging essentially the same rates so there's no real price competition happening. Short trips run about $9.50 per mile and long trips $4.43 per mile, but that gap isn't vendors doing anything clever. It's just the fixed base fare getting divided over fewer miles on a short trip compared to a longer one.

Night hours only account for 12% of total revenue which seems low. The extra surcharge already applies to 62% of trips and concentrates around 4 to 8 PM, so the evening peak gets a premium built in. The night window doesn't have anything similar, and it shows in the revenue share.

RECOMMENDATIONS

- Introduce mild surge pricing of 10 to 15 percent during peak hours (5 PM to 7 PM on weekdays) when demand consistently exceeds supply. The data shows 127,990 sampled trips in the single 6 PM hour, equivalent to 2.56 million annual trips.
- Short trips are already the highest-margin segment per mile at $9.50/mi. A minimum fare floor on rides under 1 mile would protect revenue on very short rides without meaningfully deterring passengers.
- A driver incentive for the 11 PM to 5 AM window could grow supply in this underserved segment, which currently generates only 12 percent of revenue despite strong demand in entertainment zones.
- A formalised group discount of 10 to 15 percent for trips with 3 or more passengers could stimulate demand during off-peak midday hours. At $1.29 per mile per person, groups already find taxis cost-competitive. A small discount could shift more group journeys from ride-hailing to yellow taxis.

Overall Conclusion

Stepping back, the NYC taxi network in 2023 runs on a pretty concentrated base. Around 80% of revenue comes from the 6 AM to 10 PM window, Manhattan and the two airports handle most of the trips, and over 81% of payments go through card. The behavioural finding that stood out most to me was tip percentage being inversly tied to trip length rather than fare amount. Shorter, cheaper trips actually get tipped better on a percentage basis, which isn't what you'd expect going in.

Distance is the dominant driver of fare at r = 0.9452. With both vendors charging identical rates and no real competition on price, some kind of demand-based pricing during the peak hours is probably the most direct way to grow revenue without changing anything else about how the fleet operates.

> **Final thought:** The opportunities here are pretty clear from the data. Get cabs in position before 6 PM, address the airport dead-mile problem, consider surge pricing for peak hours, and give drivers a reason to work the night shift. None of this requires a big overhaul. It just needs the data to back the decisions, which is what this analysis was trying to do.